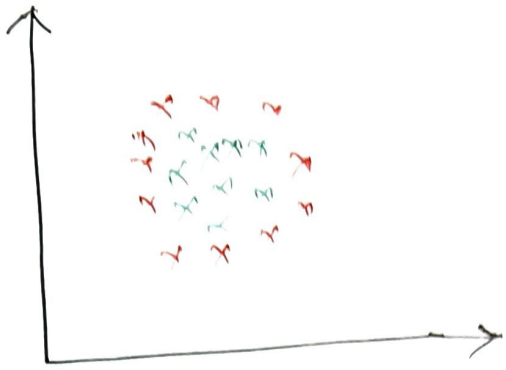
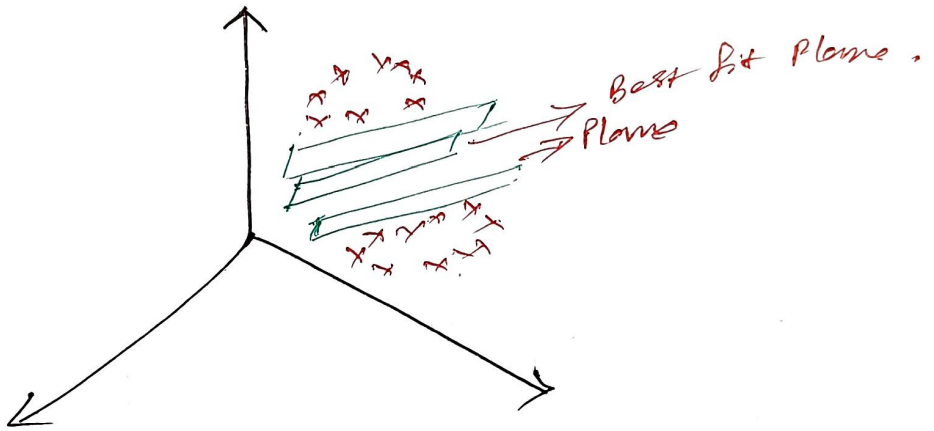
SVM Kernel



→ We can't use SVC here, result will be bad.

→ In SVM Kernel, we first convert the 2D data points to 3-D points.

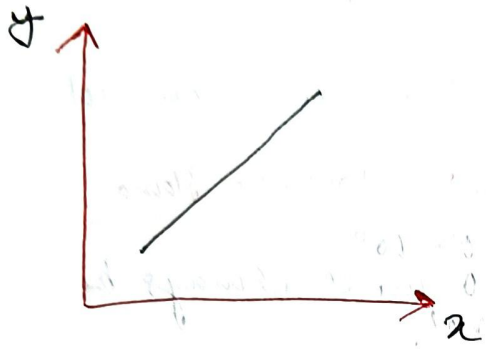
2-D $\xrightarrow{\text{Transformation}}$ 3-D



29-Oct-2024

Support Vector Machines (SVM)

- We can solve both classification and regression problem.
- classification → SVC (Support Vector classification)
  Regression → SVR (Support Vector Regression)



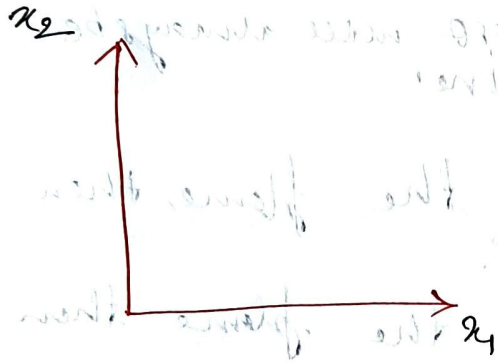
$y = mx + c$

$y = \beta_0 + \beta_1 x$

$ax + by + c = 0$

$y = \boxed{-\frac{a}{b}} x - \boxed{\frac{c}{b}}$

Coefficient, Intercept



$ax_1 + bx_2 + c = 0$

$w_1 x_1 + w_2 x_2 + b = 0$

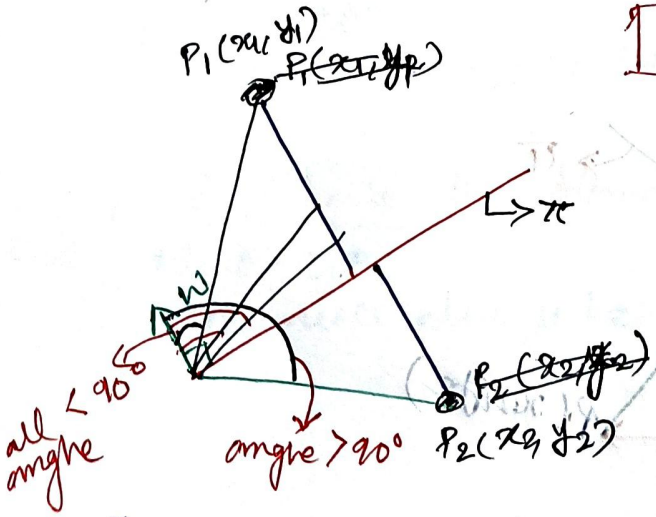
$w^T x + b = 0$ → If this line passes through origin then $b = 0$

Matrix multiplication

$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} [x_1 \; x_2]$

Eqn of line passes through origin

$\boxed{w^T x = 0}$



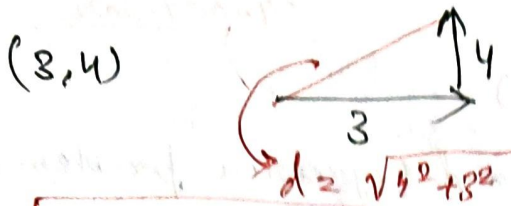
Distance of a point to the plane,

$\boxed{d = \frac{w^T p_1}{\|w\|} = \|w\| \, \|p_1\| \cos\theta}$

We need unit vector that's why divided by magnitude of w.

π = line in 2-D and plane in 3-D

w = vector



$d = \sqrt{4^2 + 3^2} = 5$

Unit vector, $\hat{d} = \frac{d}{\|d\|}$ → Magnitude

$\left(\frac{3}{5}, \frac{4}{5}\right) \rightarrow d = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = 1$

→ Unit vector is a way to get focused on direction not on magnitude.

Upward Vector

$$d = \frac{W^T P_1}{\|W\|}$$

$$d = \|W\| \|P_1\| \cos\theta$$

Point below the Plane

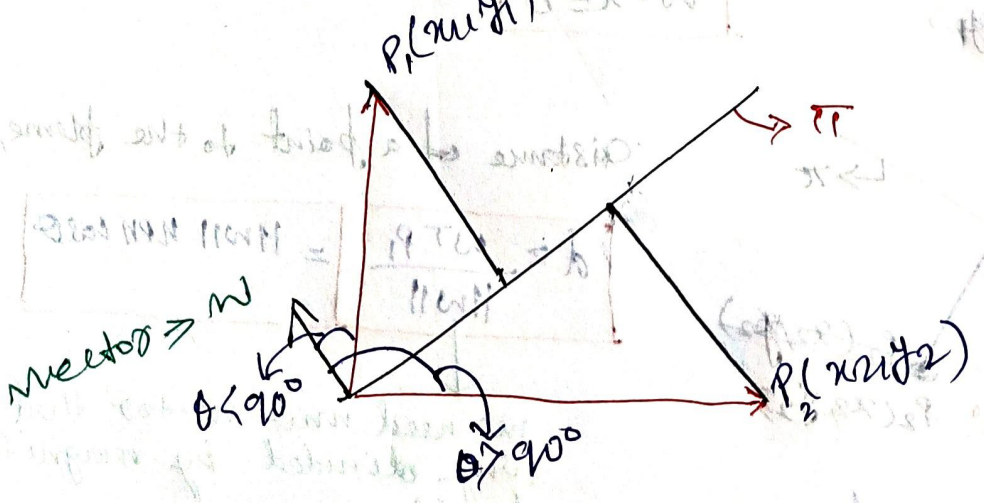
as $\theta > 90°$
$\cos\theta$ will always be (-ve).

Point above the Plane

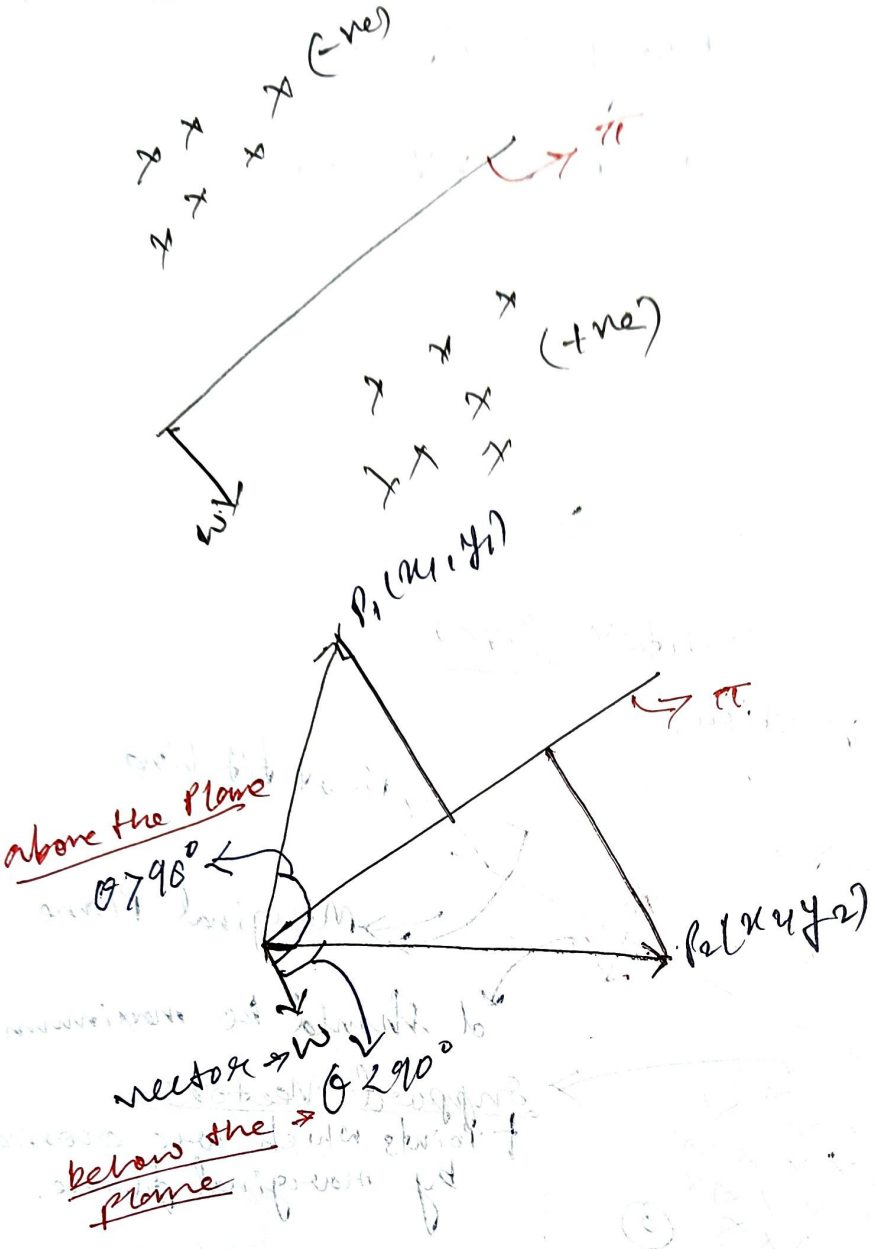
as $\theta < 90°$
$\cos\theta$ will always be (+ve)

→ If any point falling above the plane, then θ will be less than 90°.

→ If any point falling below the plane then θ will be greater than 90°.



⇒ Downward Vectors



Point below the plane

as $\theta < 90°$

$\cos\theta$ will always be '+ve'

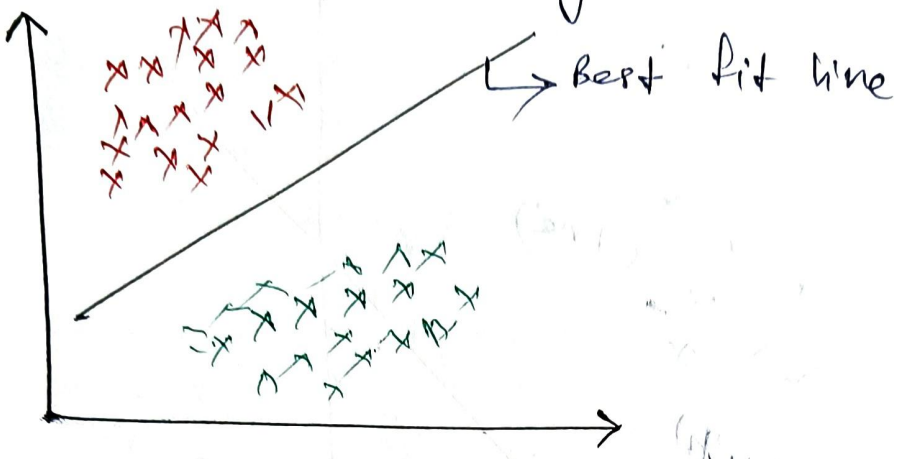
Point above the plane

as $\theta > 90°$

$\cos\theta$ will always be '−ve'

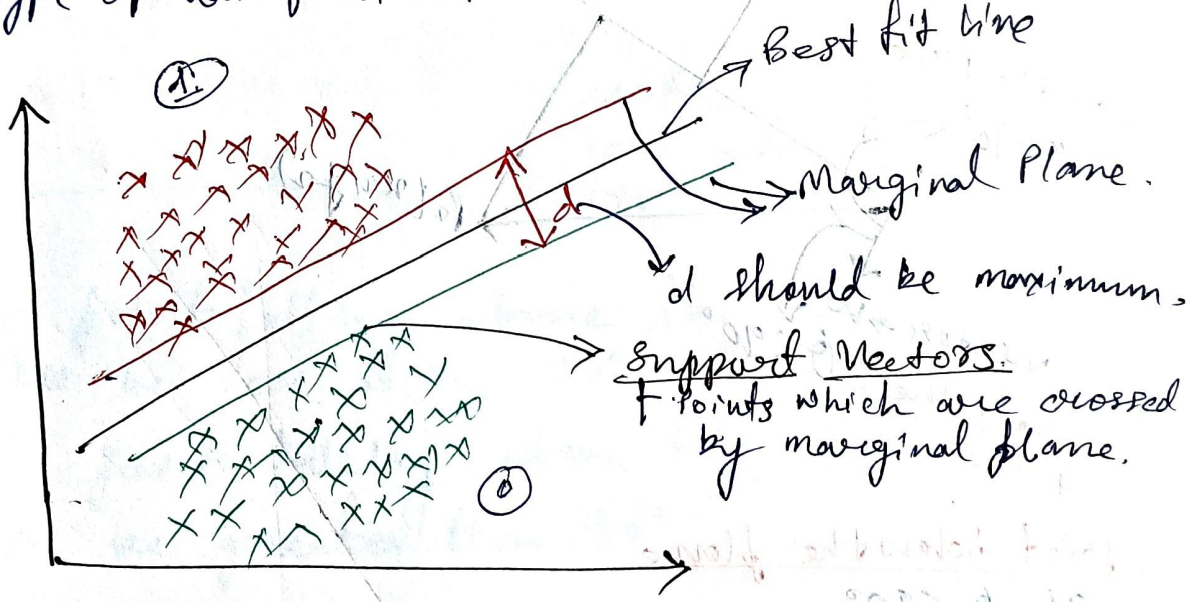
Geometric Intuition Behind Support Vector Machine

Logistic Regression (Binary classification)



Support Vector Classifier (SVC)

⇒ Any type of classification.



Marginal Plane

- line passes through nearest point.

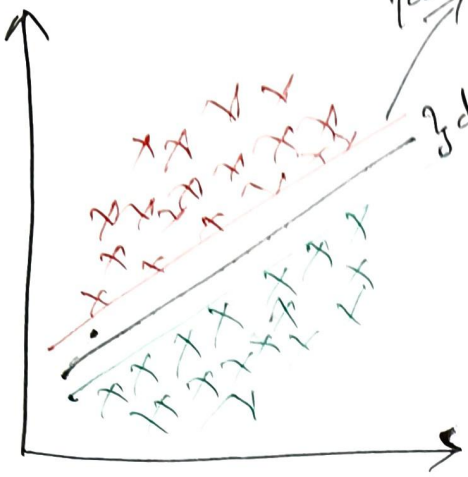
Hard Margin

- less error we will able to seperate all the points without any error.

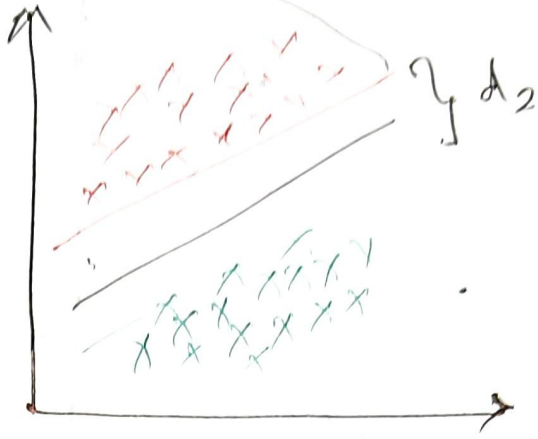
Soft Margin

- Having error. we so we'll not get marginal plane without any error.

Hard Margin (without Error)



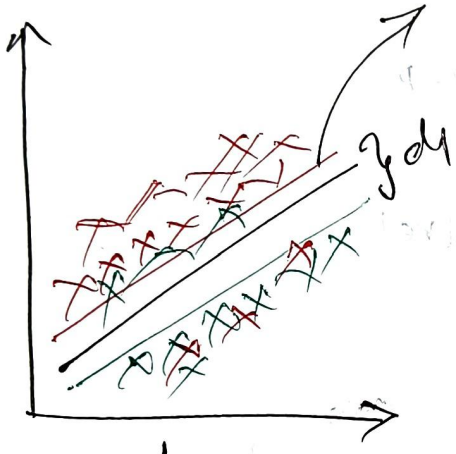
Bad



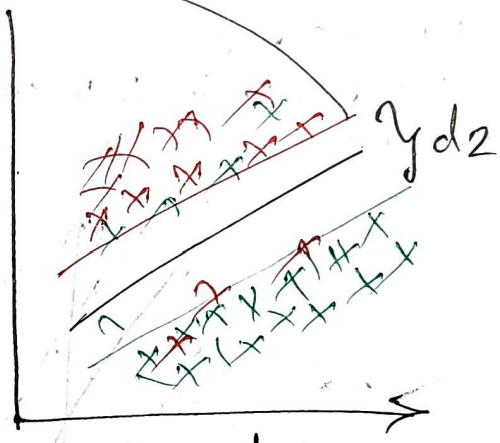
Good

$d_2 > d_1$

Soft Margin (with Error)



Bad

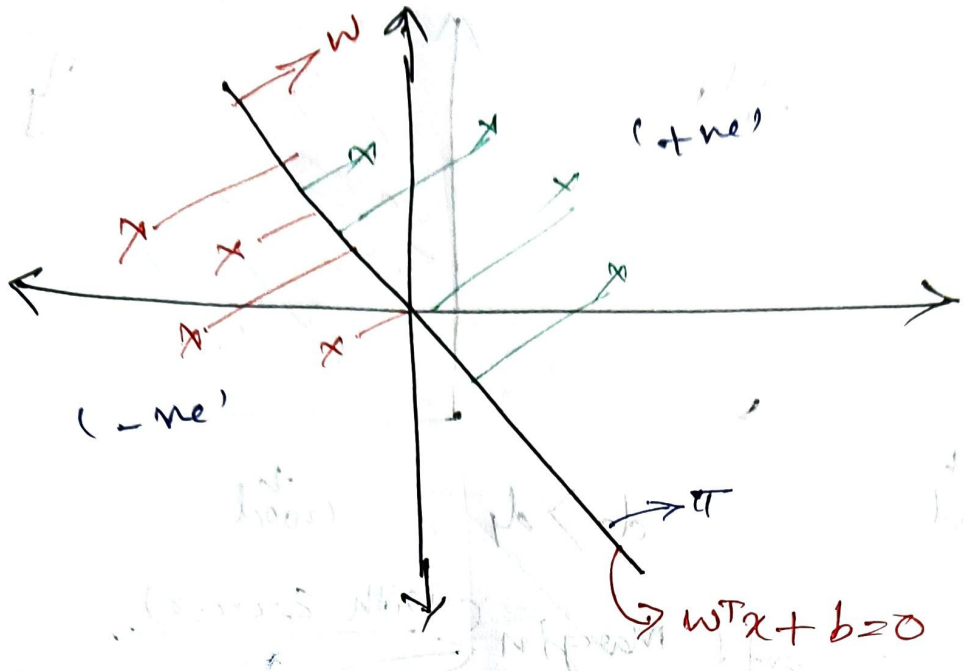


Good

$d_2 > d_1$

→ Marginal Plane should be equidistance from best fit line.

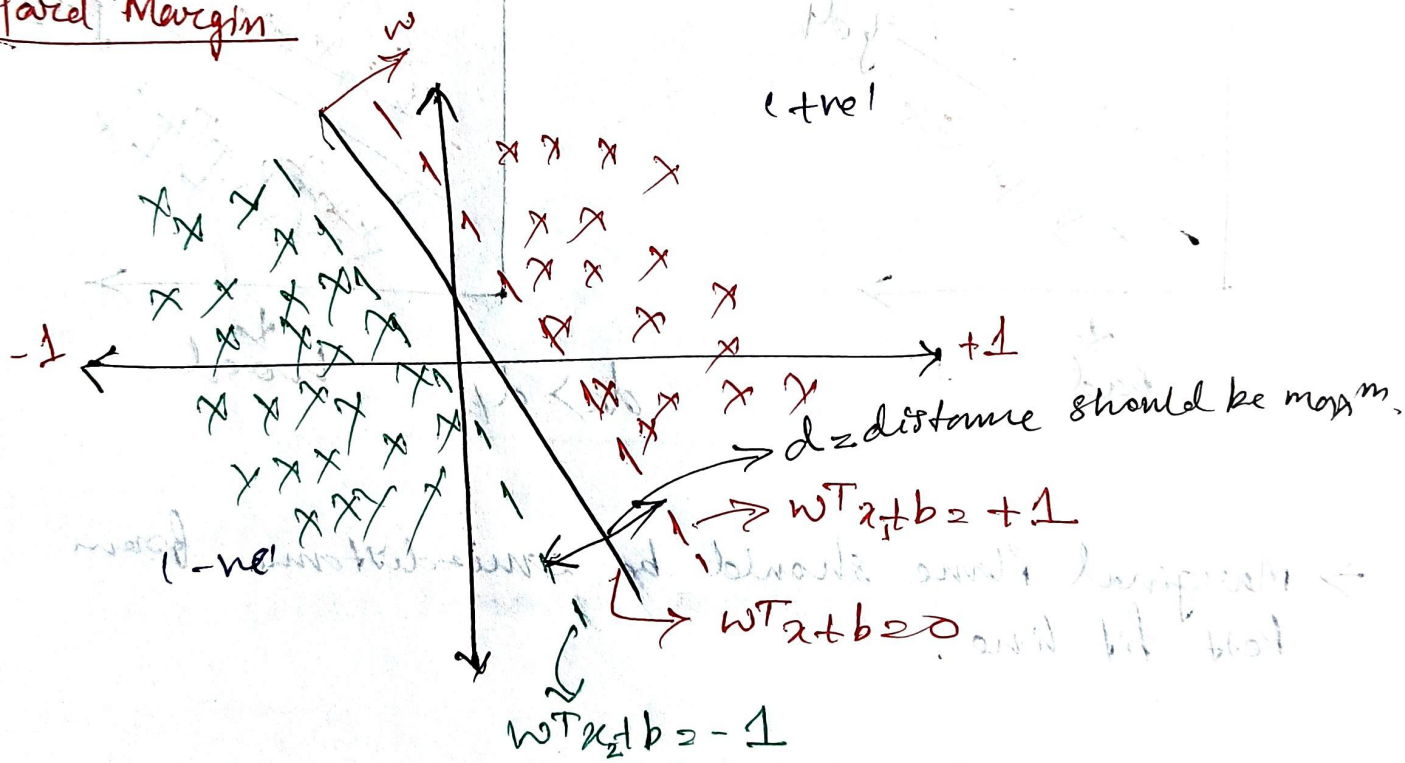
SVM Mathematical Intuition



w = vector

π = Plane in 3-D and line in 2-D

Hard Margin



$$w^T x_1 + b = +1 \quad \text{---(1)}$$

$$w^T x_2 + b = -1 \quad \text{---(2)}$$

(1) - (2) (-) (-) (+)

$$w^T (x_1 - x_2) = 2$$

Unit vector of w,

$$\frac{w^T (x_1 - x_2)}{\|w\|} = \frac{2}{\|w\|}$$

cost function

Maximize, $\frac{2}{\|w\|}$ → Distance b/w Marginal Plane
w, b

↳ change it to get maximum distance

constraint such that — condition to fit correct classified points

$y_i \begin{cases} 1 & \cdot \; w^T x + b \geq 1 \\ -1 & \cdot \; w^T x + b \leq -1 \end{cases}$

For all classified correct points

constraints → $y_i * (w^T x + b) \geq 1$

Maximize $\frac{2}{\|w\|}$ by changing w, b } Both are same.

Minimize $\frac{\|w\|}{2}$ by changing w, b

→ Loss function focus on minimization

cost function

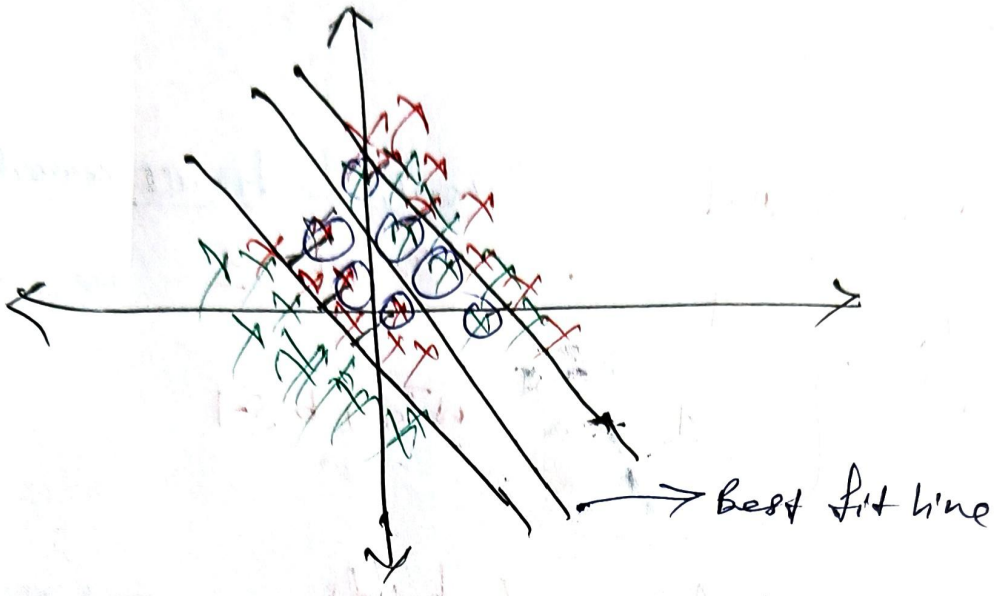
Min $\frac{\|w\|}{2}$ by changing w, b

$$\text{Min} \quad \frac{\|w\|}{2} + C_i \sum_{i=1}^{n} |\xi_i|$$

↓

Hinge loss for soft margin

$\zeta_i \geq$ How many points we can ignore for misclassification.

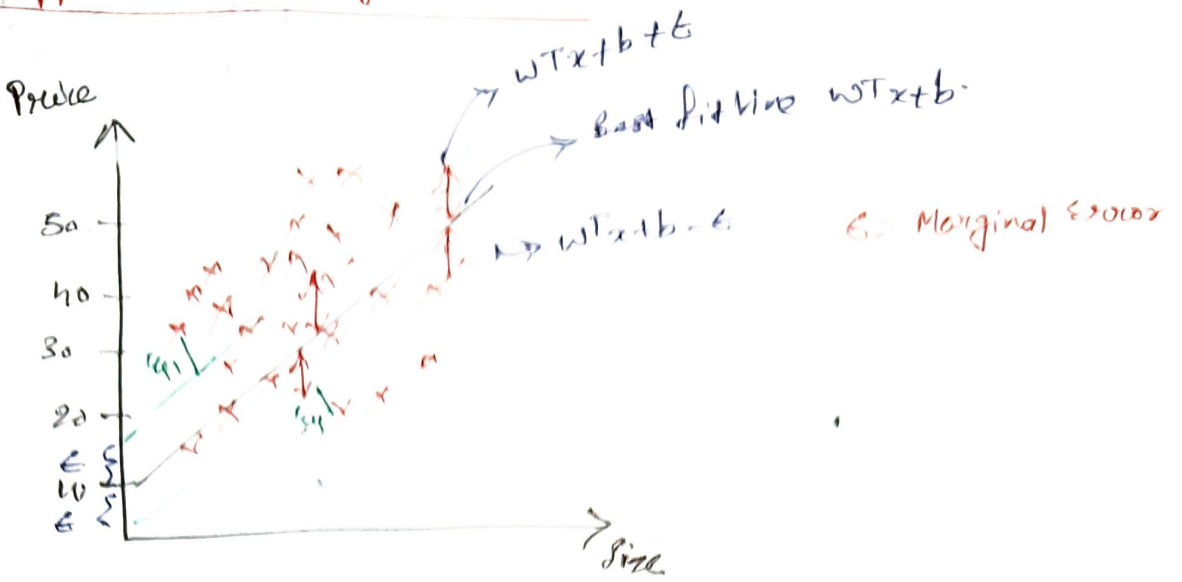


$\zeta_i \geq$ If points are miss classified → Ignore.

$\zeta_i \geq$ Summation of the distance of the incorrect data points from the marginal plane.

If points distance from the marginal plane.

SVR (Support Vector Regressor)



Cost function

Minimize by changing w, b

$$\frac{\|w\|^2}{2} + C_i \sum_{i=1} |\xi_i|$$

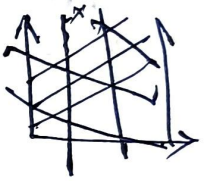
$|\xi_i|$ → Hinge loss

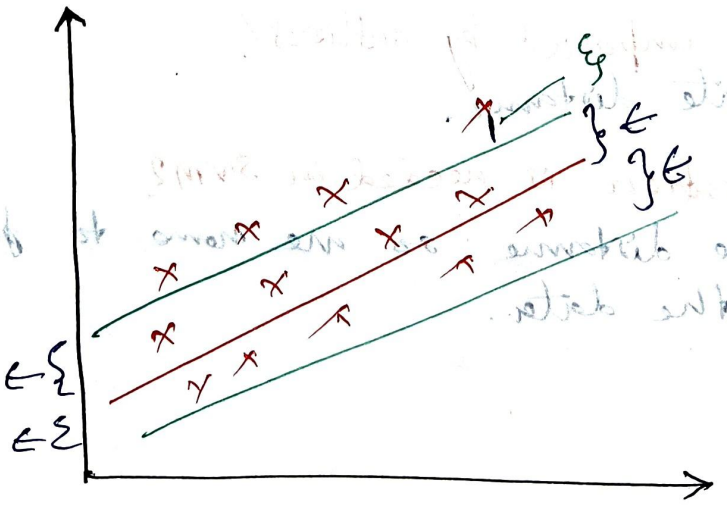
constraint

$$|y_i - w_i x_i| \leq \epsilon + |\xi_i|$$

Diff b/w tenth and the predicted point

value should be under the thus value



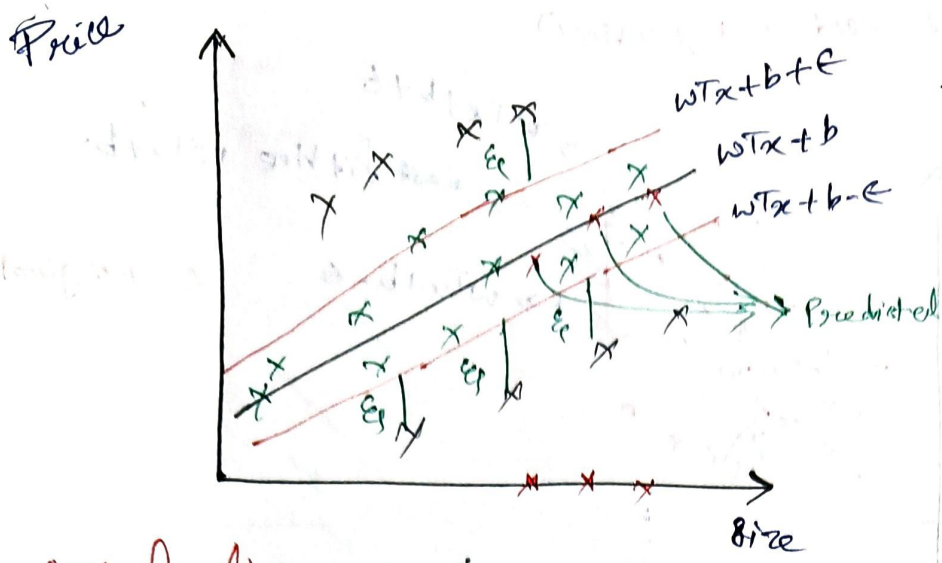


ϵ = Marginal Error to decide marginal plane

Hyperparameter

→ Keep adjusting ξ_i to get best margin.

→ We could say incorrect point in regressor.



SVR Regressor

- No compute incorrect value, because it will be continuous value.

Hyperparameter

- ϵ → Marginal Error

ξ_i → Distance of point from marginal plane

Cost Function

$$\underset{w/b}{\text{Min}} \quad \frac{\|w\|}{2} + C \sum_{i=1}^{n} |\xi_i|$$

↓

Hinge loss

Constraint

$$|y_i - w_i x_i| \leq \epsilon + |\xi_i|$$

ϵ → Margin of Error

ξ_i → Error above the margin.

Q1 Will SVM get impacted by outliers?

Ans Yes, it calculate distance.

Q2 Does Standardization is needed in SVM?

Ans Yes, it involve distance so we have to perform Scaling of the data.